

Welfare Implications of Nominal GDP Targeting in a Small Open Economy

Fabrizio Sebastian Solf Rivas*

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Abstract

Nominal GDP (NGDP) targeting rules have attracted interest in the literature as a monetary policy alternative, owing to the greater clarity they would give a central bank in explaining its policy objectives, the fewer problems involved in measuring the target variable, but mainly because of their performance in the face of shocks relative to traditional frameworks of monetary policy. Within a New Keynesian model with nominal price and wage rigidities, this paper extends the analysis of the welfare implications of NGDP rules by examining a small open economy scenario. Starting from a welfare function derived from the utility of consumers, the NGDP target is compared with a domestic inflation target, a CPI inflation target, a wage inflation target and a Taylor rule. The rule that minimises the welfare loss is identified under different calibrations, considering supply and demand shocks. The results of the simulations show the advantages that the adoption of policy rules reacting to NGDP confers on a central bank in the face of supply shocks, while in the face of demand shocks there are no material differences relative to CPI inflation targets. A final exercise extends the framework to a commodity-exporting emerging economy, where NGDP targeting remains preferable to CPI targeting provided that wage rigidities are at least as large as price rigidities.

Keywords: small open economy, monetary policy, nominal GDP targeting, optimal monetary policy, DSGE model.

JEL codes: E32, E52, F31, F41, F43.

*Universidad del Pacífico, Lima, Peru. This paper builds on the author's undergraduate research work in Economics (Universidad del Pacífico, 2022). All errors are my own.

1 Introduction

Monetary policy plays an important role in maintaining stability in the economy and the welfare of citizens. The experience of small open economies over the last decades bears this out: a good number of them have preserved macroeconomic stability owing, in part, to the sound performance of their central banks.

In the aftermath of the COVID-19 pandemic, the traditional policy rules applied by central banks around the world were put to the test, with weakened economies and circumstances that narrowed the scope of action of monetary policy. Moreover, in a context of climate change and supply chain disruptions, countries are expected to face supply shocks more frequently, especially those that depend more heavily on their exports.

Supply shocks, unlike demand shocks, cannot be fully controlled by central banks, and may end up causing deviations from inflation targets. This is illustrated by the record of small open economies operating under inflation targeting, where a sizeable share of the deviations from the target range originates in supply shocks.

In that sense, it is necessary to evaluate the properties and benefits of alternative rules, above all for small economies, whether industrialised or not. Among the unconventional policies, nominal GDP (NGDP) targets have generated interest in recent years owing to their potential benefits over traditional policy rules when reacting to different shocks.

NGDP targeting is an idea initiated by Meade (1978) and Tobin (1980), and subsequently taken up by Sumner and Roberts (2018), though from the perspective of securing employment. Successive crises have modified the requirements placed on policy rules as a result of the current conditions of the economy. NGDP targeting shares the features of inflation targeting, but is implicitly aimed at both inflation and real output, while it should provide the ability to handle zero lower bound scenarios. For both developing and developed countries it is an interesting alternative, since it deals pre-emptively with acute crises such as that of 2008 and with supply shocks such as those of 2022.

This said, the objective of this paper is to study the desirability of an NGDP target within the context of a New Keynesian model with price and wage rigidities in a small open economy. Specifically, we compare it with the usual rules on the basis of the computed social welfare loss that each rule generates when the economy faces exogenous shocks.

Unlike previous research, this paper contributes to the literature by analysing under which type of monetary policy rule the minimum welfare loss arises in a small open economy. By simulating the model under different states of the economy, we seek to show that an NGDP targeting rule generates lower welfare losses in the face of supply shocks, while in the face of demand shocks it displays a performance similar to that of inflation targeting (IT).

The approach of the paper is mainly theoretical, in the sense that it seeks to define the conditions under which NGDP is an optimal rule in the open economy model set out below. To that end, this policy rule is analysed under different parameterisations, such as the degree of openness and the level of the rigidities. Optimal rules are computed by numerical methods

and welfare comparisons are then carried out. In addition, the desirability of NGDP is assessed for a developing economy; for this purpose the model is calibrated with parameters drawn from the literature that replicate the basic features of such an economy. On the basis of the results obtained, we explain the transmission mechanism that allows an NGDP rule to deliver optimal outcomes.

The paper is organised as follows. Section 2 presents a review of the literature relevant to the different traditional policy measures, in order to compare them with NGDP monetary policy. Section 3 presents a standard New Keynesian open economy model with rigidities, together with the monetary policy equations to be compared. Section 4 clarifies the methodology, in which the optimal rule is determined under different calibrations by means of a welfare function, from which impulse response functions are obtained. Section 5 reports the results of the simulations for the different shocks analysed, after which robustness tests are carried out. Finally, Section 6 presents the conclusions and recommendations for extending the present work.

2 Literature review

Monetary policy is a tool whose objective is to reduce the volatility of business cycles arising from different shocks, in such a way that it allows the welfare of the population to be optimised. The mandate that each monetary authority maintain stable inflation, or also achieve employment, is crucial for economic agents. In this way, output is made to grow around its long-run trend. The aim is to attain credibility and sound social indicators, to raise investment and to avoid negative consequences for the economy. Policymakers use monetary rules, which consist of instruments that respond to different states of the economy in order to mitigate shocks. The presence of rigidities, imperfections and shocks of different natures generates trade-offs, which creates the need for an analysis of optimal monetary policy.

2.1 Monetary policy in New Keynesian models

The literature generally conducts monetary policy analysis under New Keynesian models, because of their ability to perform comparisons across different monetary policy rules. Clarida et al. (1999) present an analysis of monetary policy for a closed economy with nominal rigidities showing that optimal monetary policy incorporates an inflation target rule in order to reduce volatility over time. The optimal rule is efficient in perfectly accommodating demand shocks; in the face of supply shocks, however, it presents a trade-off between inflation and economic activity (the output gap).

Galí and Monacelli (2005) extend the analysis by studying monetary policy in a New Keynesian model with an open economy and price rigidities. This model incorporates a trade-off between stabilising the nominal exchange rate and the terms of trade, in addition to stabilising domestic inflation and the output gap. In order to determine the optimal monetary policy, they show analytically that, in the particular case of a logarithmic utility function and unit elasticity

of substitution, a rule that strictly minimises domestic inflation is the optimal monetary policy.

Despite their simplicity, the previous models show that monetary policy cannot be understood as a tool capable of fully stabilising fluctuations. The inclusion of additional rigidities such as wages (Erceg et al. 2000) adds new scenarios that must be considered in order to maintain stability in an open economy, as demonstrated by Campolmi (2014), where the optimal rule ceases to be a domestic inflation target.

2.2 Traditional inflation rules

Central banks tend to share a consensus in favour of monetary rules rather than discretion. The latter leads agents to inaccuracies in their expectations, which worsens the trade-off between inflation and the output gap that traditional monetary policies usually face. Moreover, for the public the objectives of the bank are simple to understand under a monetary policy rule, because, given that the central bank is bound to comply with it, expectations can be formed. There is also the factor of frictions in the measurement of variables (Garín et al. 2016).

The traditional inflation rules that satisfy the characteristics mentioned above are the Taylor rule, strict domestic inflation targeting and strict consumer price index (CPI) targeting. The participation of these rules in emerging economies produced improvements, not without also generating criticism of their performance. According to Frankel (2010), after the emergence of inflation targeting (IT) in 1971, following the abandonment of fixed exchange rates, emerging economies that adopted it enjoyed declines in inflation and lower growth volatility. Nevertheless, after the crisis of 2008 other types of anchors were proposed, such as a CPI target, in order to pay attention to other nominal variables such as asset prices, exchange rates and commodity prices, given that these are predictors of future inflation.

Inflation targeting was designed for developed economies and is not as well suited to emerging countries (for example, small open economies), since exogenous shocks are not usually considered together with poor conditions in the external accounts or with the need to finance the trade deficit, on the assumption that international markets should smooth consumption. According to Frankel, in emerging countries the volatility of supply shocks is greater (commodity or energy prices). Booms feature capital inflows, currency overvaluation and associated current account deficits, and are then followed by crashes, abrupt depreciation and recession. After experiencing such scenarios, it was considered that the weight of nominal GDP should be split between inflation and output, aiming at a “core” CPI target instead of the CPI alone; but a core CPI inflation target is complicated to explain and undermines credibility, which is even worse in emerging countries (Frankel 2010).

According to Céspedes et al. (2014), the use of non-traditional policies during the crisis of 2008 showed that central banks had the ability to counteract external shocks and to keep recessions short and shallow. Monetary and exchange rate policies thus deviate from the textbook IT paradigm.

2.3 Nominal GDP targeting

The literature presents nominal GDP targeting (NGDP) as a policy alternative to be taken into account because of its possible advantages over inflation rules or the Taylor rule. First, it is noted that an NGDP rule is easier to explain to the public and makes it easier for the central bank to follow rules under commitment (McCallum 2015; Sumner 2014). Second, unlike regimes that react to inflation, NGDP is more efficient at stabilising cost shocks (Garín et al. 2016). Third, NGDP also reacts to inflation and output, but has the advantage that it responds to a single variable (nominal output is a combination of both) (Chen 2021). Finally, NGDP is easier for the central bank to observe than the output gap, which consists of an estimate and therefore embeds a larger margin of error that may translate into less efficient policy responses (Beckworth and Hendrickson 2020).

Garín et al. (2016) evaluate through simulations the properties of NGDP, in levels (NGDP-LT) and in growth rates (NGDP-GT), in a New Keynesian closed economy model, first with price and wage rigidities. By means of a function that measures the welfare loss arising from different shocks, the authors compare the NGDP-GT rule with strict inflation targeting, strict output gap targeting, and a conventional Taylor rule. The results indicate that NGDP-GT dominates inflation targeting and the Taylor rule, above all in the face of supply shocks.

Chen (2021) extends the results in a model with growth and productivity trends, in the presence of supply and demand shocks. He compares inflation targeting with NGDP-LT and NGDP-GT. The simulations, subject to different shocks and conditions, show that the NGDP-GT rule generates a lower welfare loss relative to the rest.

The literature also points out that taking into account the measurement problem in the output gap variable confers advantages on an NGDP rule. Garín et al. (2016) evaluate the issue through a model that includes a noise variable in the output gap, and determine that, given a certain threshold of noise in the variables, an NGDP-GT target is preferable. The implications related to the correct measurement of the target variable are explored by Beckworth and Hendrickson (2020), who show that, under imperfect information, one cannot consider an NGDP rule as a special case of the Taylor rule, because the output gap is subject to larger measurement errors in real time. They therefore argue that a proper comparison between the two rules must consider the information problem, and by means of simulations with measurement error in the output gap they determine that an NGDP rule is preferable to the Taylor rule.

The intuition behind the advantage of an NGDP target, as regards the way it operates, is related to its ability to respond to two variables (or a dual mandate) through a single variable. By construction, the NGDP target is composed of inflation and real GDP, and in that sense it has a structure similar to a Taylor rule, with the advantage that it does not suffer from the information problem of the output gap. It is for this reason that, in the face of demand shocks, NGDP should generate a response similar to inflation targets.

The main advantage lies in its response to supply shocks, where NGDP produces smaller welfare losses. In the face of these shocks, inflation targets focus only on reducing price volatility, which is suboptimal. NGDP, for its part, assigns the same weight to both deviations, so

that a smoother response to supply shocks is delivered, which benefits the real sector of the economy.

One way of seeing how NGDP interacts is through the balance sheet mechanism operating via the instruments of a policymaker. The policymaker increases the money supply, this raises the price of assets and reduces adverse selection and moral hazard, which are usual in the face of negative supply shocks. As the value of collateral falls, the probability of losses from choosing clients with riskier projects increases (adverse selection). On the other hand, in the face of the fall in net worth, reducing the participation of the firm's shareholders in the projects they undertake may generate the incentives to carry out riskier projects (moral hazard) (Mishkin 1995). NGDP thus increases lending, and this causes investment to rise so that output grows, offsetting the fall caused by the negative supply shock.

From another point of view, increasing the money supply also causes the interest rate to fall, so that cash flows increase. This is why adverse selection and moral hazard also diminish, with the result that lending does not fall as much and investment is sustained even in scenarios in which output is vulnerable.

3 Analytical framework

To determine the type of monetary policy rule that is optimal from a welfare loss minimisation perspective, we take as our basis the New Keynesian model for a small open economy proposed by Campolmi (2014), which includes price and wage rigidities. Certain modifications are then made to the model in order to adapt it to the requirements of NGDP-GT (henceforth referred to simply as NGDP) following Garín et al. (2016).

The use of the analytical framework presented is useful because it allows different dynamics to be recognised, which are relevant when defining optimal monetary policy. We currently have a trade-off between the control of the exchange rate and inflation (open economy), and a trade-off between stabilising the real side of the economy and inflation (nominal rigidities). The latter is an important factor, given that a main reason why NGDP is an interesting alternative is its ability to respond to shocks taking into account that these two objectives, prices and the real economy, must be balanced. As will be shown below, this is achieved because wage rigidities amplify the effect of inflation on the economy and generate additional distortions to prices.

The model features the following agents: households, intermediate goods producing firms, and final goods producing firms.

3.1 Households

The economy is inhabited by a representative household that seeks to maximise the following intertemporal utility function:

$$E_0 \sum_{t=0}^{\infty} \beta^t U(C_t, N_t(h), Z_t), \quad (1)$$

where $N_t(h)$ denotes hours of work for the different types of labour, Z_t is an exogenous preference shifter, and β is a discount factor. C_t is a consumption index composed of the consumption of domestic goods ($C_{H,t}$) and the consumption of foreign goods ($C_{F,t}$), which takes the following form:

$$C_t = \left[\alpha^{\frac{1}{\eta}} C_{H,t}^{\frac{\eta-1}{\eta}} + (1-\alpha)^{\frac{1}{\eta}} C_{F,t}^{\frac{\eta-1}{\eta}} \right]^{\frac{\eta}{\eta-1}}, \quad (2)$$

where $\eta > 0$ measures the degree of substitution between goods from different countries, and α measures the degree of home bias (if $\alpha = 1$ we have the closed economy case). Aggregate consumption takes the following form:

$$C_{H,t}(j) = \left[\frac{P_{H,t}(j)}{P_{H,t}} \right]^{-\varepsilon_p} C_{H,t}, \quad C_{i,t}(j) = \left[\frac{P_{i,t}(j)}{P_{i,t}} \right]^{-\varepsilon_p} C_{i,t}, \quad (3)$$

$$C_{i,t} = \left[\frac{P_{i,t}}{P_{F,t}} \right]^{-\gamma} C_{F,t}. \quad (4)$$

The equations in (3) represent the demand for local good j (or for that of sector j), and the demand for good j from country i , respectively. The parameter ε_p is the elasticity of substitution between different consumption goods in the same market. Equation (4) gives the demand for the total goods of country i as a function of foreign (or imported) consumption, where γ represents the degree of substitution between foreign goods. From the cost minimisation problem we obtain the following demands as a function of total consumption:

$$C_{H,t} = \alpha \left(\frac{P_{H,t}}{P_t} \right)^{-\eta} C_t, \quad C_{F,t} = (1-\alpha) \left(\frac{P_{F,t}}{P_t} \right)^{-\eta} C_t, \quad X_t = (1-\alpha^*) \left(\frac{P_{H,t}}{P_{F,t}} \right)^{-\eta^*} C_t^*. \quad (5)$$

From these demands we arrive at an expression for the price index (or CPI), which aggregates the prices of local goods ($P_{H,t}$) and of foreign or imported goods ($P_{F,t}$):

$$P_t = \left[\alpha P_{H,t}^{1-\eta} + (1-\alpha) P_{F,t}^{1-\eta} \right]^{\frac{1}{1-\eta}}. \quad (6)$$

On the basis of these equations, the budget constraint of households is determined:

$$P_t C_t + E_t[\Omega_{t,t+1} D_{t+1}] \leq D_t + W_t(h) N_t(h) + \Pi_t, \quad (7)$$

where $\Omega_{t,t+1}$, D_t , Π_t and $W_t(h)$ are the stochastic discount factor, the payoff of the bond, the income of firms and wages, respectively. The stochastic discount factor is equal to the inverse of the interest rate on a bond ($1/R_t$). The utility function is specified as follows:

$$U(C_t, N_t(h), Z_t) = \left(\frac{C_t^{1-\sigma}}{1-\sigma} - \frac{N_t(h)^{1+\varphi}}{1+\varphi} \right) Z_t, \quad (8)$$

where σ is the coefficient of relative risk aversion and φ is the inverse of the Frisch elasticity of labour supply. Optimising (1) together with (8) with respect to consumption, subject to (7),

yields the Euler equation:

$$C_t^{-\sigma} = \beta E_t \left[C_{t+1}^{-\sigma} \frac{Z_{t+1}}{Z_t} \frac{1}{\Omega_{t,t+1}} \frac{P_t}{P_{t+1}} \right] = \beta E_t \left[C_{t+1}^{-\sigma} \frac{Z_{t+1}}{Z_t} \frac{R_t}{1 + \pi_{t+1}} \right], \quad (9)$$

where $1 + \pi_{t+1} = P_{t+1}/P_t$.

3.1.1 The wage decision

Households (h) supply their differentiated labour to the business sector (j), and hold a certain degree of bargaining power. The demand of the firm in sector j is obtained by minimising wage costs:

$$N_t(h) = N_t \left[\frac{W_t(h)}{W_t} \right]^{-\varepsilon_w}, \quad (10)$$

where ε_w is the elasticity of substitution between the different types of labour, N_t aggregates the entire labour supply, and W_t is an index that aggregates all wages. Following Erceg et al. (2000), only a fraction $(1 - \theta_w)$ can change its wage optimally so as to maximise utility, while the fraction θ_w keeps its wage constant, $W_t(h) = W_{t-1}(h)$. Substituting these conditions into the utility function and the budget constraint, we obtain:

$$\begin{aligned} \mathcal{L} = E_t \sum_{s=0}^{\infty} (\beta \theta_w)^s \left\{ - \left(\frac{\left(\frac{w_t(h) \pi_{t,t+s}^{-1}}{w_{t+s}} \right)^{-\varepsilon_w(1+\varphi)} N_{t+s}^{1+\varphi}}{1 + \varphi} \right) Z_{t+s} \right. \\ \left. + \lambda_{t+s} P_{t+s} \left[w_t(h) \pi_{t,t+s}^{-1} \left(\frac{w_t(h) \pi_{t,t+s}^{-1}}{w_{t+s}} \right)^{-\varepsilon_w} N_{t+s}^{1+\varphi} \right] \right\}. \end{aligned} \quad (11)$$

Equation (11) is the household's Lagrangian in real terms under Calvo-type wage rigidities, where we display only the variables related to wages; w_t denotes the real wage W_t/P_t , and $\pi_{t,t+s}$ denotes cumulative inflation s periods ahead. In this optimisation problem, in each period the household faces a probability θ_w of being unable to update its wage, so that this acts as a discount factor additional to β . The optimal wage in t can be expressed as follows:

$$w_t^{\#, 1+\varepsilon_w \varphi} = \frac{\varepsilon_w}{\varepsilon_w - 1} \frac{E_t \sum_{s=0}^{\infty} (\beta \theta_w)^s w_{t+s}^{\varepsilon_w(1+\varphi)} \pi_{t,t+s}^{\varepsilon_w(1+\varphi)} N_{t+s}^{1+\varphi} Z_{t+s}}{E_t \sum_{s=0}^{\infty} (\beta \theta_w)^s C_{t+s}^{-\sigma} Z_{t+s} w_{t+s}^{\varepsilon_w} \pi_{t,t+s}^{\varepsilon_w-1} N_{t+s}}. \quad (12)$$

Since we know the wage that is updated in each period, equation (12), we can determine the dynamics of wages. This equation indicates that the aggregate wage is a weighted average of the updated wage and the sticky wage. It is important to note that, in the equation for the dynamics of the real wage, the general price level (P_t) affects the evolution of the wages of those who fail to re-optimize:

$$w_t^{1-\varepsilon_w} = (1 - \theta_w) w_t^{\#, 1+\varepsilon_w \varphi} + \theta_w (1 + \pi_{t+1})^{\varepsilon_w-1} w_{t-1}. \quad (13)$$

In the special case in which there are no sticky wages ($\theta_w = 0$), the optimal wage would be $w_t = \frac{\varepsilon_w}{\varepsilon_w - 1} \frac{N_t^\phi}{C_t^{1-\sigma}}$, where there is a mark-up over the marginal rate of substitution between consumption and labour. In the case in which $\varepsilon_w \rightarrow \infty$, that is, there is full substitutability between the different types of labour (they are not differentiated at all), we obtain the flexible wage case of Galí and Monacelli (2005): $w_t = C_t^\sigma N_t^\phi$.

3.1.2 Real exchange rate and terms of trade

The terms of trade are given by S_t , defined as the ratio of the price of foreign goods to the price of domestic goods: $S_t = P_{F,t}/P_{H,t}$. We assume that the law of one price holds. The expression $P_t = E_{i,t}P_t^i$ represents the bilateral nominal exchange rate with country i . The real exchange rate, which is the ratio of the foreign to the domestic consumer price index, assumed to be expressed in units of the domestic currency, is

$$Q_{i,t} = \frac{E_{i,t}P_t^i}{P_t}. \quad (14)$$

3.1.3 International risk sharing

Under the assumption of a complete set of internationally traded state-contingent assets, the Euler equation must hold for the remaining households in the world. Thus, given an identical utility function, it takes the following form:

$$\beta E_t \left[\left(\frac{C_{t+1}}{C_t} \right)^{-\sigma} \frac{Z_{t+1}}{Z_t} \frac{1}{\Omega_{t,t+1}} \frac{P_t}{P_{t+1}} \right] = \beta E_t \left[\left(\frac{C_{t+1}^i}{C_t^i} \right)^{-\sigma} \frac{Z_{t+1}}{Z_t} \frac{1}{\Omega_{t,t+1}} \frac{E_{i,t}P_t}{E_{i,t+1}P_{t+1}} \right].$$

3.2 Firms

The supply side of the economy is divided in two: intermediate goods producing firms and final goods producing firms, whose output aggregates the inputs produced by the intermediate goods firms, which are imperfect substitutes. There is a market of monopolistic competition among intermediate firms to the extent that the goods produced are differentiated, which confers market power upon them. However, they cannot freely adjust their price in each period, but rather do so according to a stochastic process (à la Calvo).

3.2.1 Final goods firms

In the final goods market firms operate under perfect competition. The input they use is the intermediate good j . From the profit maximisation problem we obtain the demand for this good j , which depends on the relative price of intermediate good j , the elasticity of substitution between intermediate goods, and a share of aggregate output Y_t :

$$Y(j) = \left[\frac{P_{H,t}(j)}{P_{H,t}} \right]^{-\varepsilon_p} Y_t. \quad (15)$$

3.2.2 Intermediate goods firms

A typical firm in the domestic economy produces a differentiated good with the following production function and technology:

$$Y_t(j) = A_t N_t(j). \quad (16)$$

Labour is required, so that the firm faces a common wage as a cost. Firms choose their labour demand by minimising costs given their level of technology, which results in an expression for the real marginal cost: $mc_t = \frac{W_t}{A_t P_{H,t}}$.

3.2.3 Price setting

Similarly to the case with nominal sticky wages, in each period a firm has a probability $1 - \theta_p$ of changing its price optimally, while it retains the price of the previous period with probability θ_p . Given this, in each period t the firm has to take this probability into account when discounting its revenues. This can be represented as

$$\max E_t \sum_{s=0}^{\infty} \Lambda_{t+s} [P_{H,t+s}(j) Y_{t+s}(j) - W_{t+s} N_{t+s}(j)], \quad (17)$$

where $\Lambda_{t+s} = (\beta \theta_p)^s \frac{U'(C_{t+s})}{U'(C_t)}$ is a stochastic discount factor. From (17) we can arrive at the equation that describes the dynamics of prices (algebraic manipulations are performed on the first order conditions):

$$P_{H,t}^{\#} = \frac{\varepsilon_p}{\varepsilon_p - 1} \frac{E_t \sum_{s=0}^{\infty} \Lambda_{t+s} mc_{t+s} Y_{t+s} P_{H,t+s}^{\varepsilon_p}}{E_t \sum_{s=0}^{\infty} \Lambda_{t+s} Y_{t+s} P_{H,t+s}^{\varepsilon_p - 1}}. \quad (18)$$

In the special case in which prices are flexible ($\theta_p = 0$), the previous expression can be re-expressed as $P_{H,t} = \frac{\varepsilon_p}{\varepsilon_p - 1} \frac{W_t}{A_t}$; that is, it depends on the nominal marginal cost times the mark-up.

On the basis of (18), we can represent the dynamics of the evolution of aggregate prices as

$$P_{H,t}^{1-\varepsilon_p} = (1 - \theta_p) P_{H,t}^{\#, 1-\varepsilon_p} + \theta_p P_{H,t-1}^{1-\varepsilon_p}. \quad (19)$$

Dividing (19) by $P_{H,t}^{1-\varepsilon_p}$ we can arrive at the dynamics of inflation:

$$(1 + \pi_{H,t})^{1-\varepsilon_p} = (1 - \theta_p) \left(\frac{P_{H,t}^{\#}}{P_{H,t}} \right)^{-\varepsilon_p} + \theta_p. \quad (20)$$

3.3 Equilibrium conditions

In equilibrium, domestic production must satisfy domestic and foreign demand (exports):

$$Y_t = C_{H,t} + X_t. \quad (21)$$

On the other hand, equilibrium in the intermediate goods market requires that supply meet demand, and together with (21) we can arrive at an expression for price dispersion:

$$\Delta_{p,t} = (1 - \theta_p) \left(\frac{P_{H,t}^\#}{P_{H,t}} \right)^{-\varepsilon_p} + \theta_p (1 + \pi_t)^{\varepsilon_p} \Delta_{p,t-1}. \quad (22)$$

Finally, equilibrium in the labour market requires supply and demand to be equal, for which it is necessary to determine the following expression for wage dispersion:

$$\Delta_{w,t} = (1 - \theta_w) \left(\frac{w_t^\#}{w_t} \right)^{-\varepsilon_w(1+\varphi)} + \theta_w \left((1 + \pi_{t+1}) \frac{w_t}{w_{t-1}} \right)^{\varepsilon_w-1} \Delta_{w,t-1}. \quad (23)$$

3.4 Exogenous variables

The model features four exogenous processes: a shock to household preferences, Z_t ; a shock to the productivity of firms, A_t ; an external demand shock, C_t^* ; and a cost (or mark-up) shock, μ_t . These shocks are assumed to follow AR(1) processes:

$$\begin{aligned} \ln(Z_t) &= \rho_Z \ln(Z_{t-1}) + \sigma_Z \varepsilon_{Z,t}, \\ \ln(A_t) &= \rho_A \ln(A_{t-1}) + \sigma_A \varepsilon_{A,t}, \\ \ln(C_t^*) &= \rho_A^* \ln(C_{t-1}^*) + \sigma_C^* \varepsilon_{C^*,t}, \\ \ln(\mu_t) &= (1 - \rho_\mu) \ln(\mu_{ss}) + \rho_\mu \ln(\mu_{t-1}) + \sigma_\mu \varepsilon_{\mu,t}. \end{aligned} \quad (24)$$

3.5 Central bank and monetary policy

On the side of the central bank, the monetary policy rules considered for evaluation are collected in Table 1.

Table 1: Simple monetary policy rules

Policy rule	Equation
Taylor rule	$i_t = \rho + \phi_{\pi_H}(\pi_{H,t} - \pi_{H,ss}) + \phi_x x_t, x_t$ the output gap
Strict CPI inflation targeting	$\pi_t = 0$
IT – PPI	$i_t = \rho + \phi_{\pi_H}(\pi_{H,t} - \pi_{H,ss})$
IT – CPI	$i_t = \rho + \phi_{\pi}(\pi_t - \pi_{ss})$
NGDP targeting	$i_t = \rho + \phi_g(g_t - g_{ss}), g_t = \pi_{H,t} \left(\frac{y_t}{y_{t-1}} \right)$
Wage inflation targeting	$i_t = \rho + \phi_W(\pi_{W,t} - \pi_{W,ss})$

In addition, we evaluate the optimal monetary policy rule derived by Campolmi (2014) for the case of a small open economy with price and wage rigidities:

$$\begin{aligned} \lambda_w \varepsilon_p \pi_{H,t} - \lambda_w (x_{t-1} - x_t) + \beta \varepsilon_w (\pi_{w,t} - E_t[\pi_{w,t+1}]) + \varepsilon_w (\pi_{w,t} - \pi_{w,t-1}) + \lambda \varepsilon_w \pi_{w,t} \\ + (1 + \lambda + \beta)(x_t - x_{t-1}) + \beta (x_t - E_t[x_{t+1}]) - (x_{t-1} - x_{t-2}) = 0, \end{aligned} \quad (25)$$

where $\lambda = \frac{(1 - \beta \theta_p)(1 - \theta_p)}{\theta_p}$ and $\lambda_w = \frac{1 - \theta_w}{\theta_w} \frac{1 - \beta \theta_w}{1 + \varphi \theta_w}$. This rule is taken as a benchmark, since, given its structure, it is difficult to estimate in practice.

3.6 The rigidity channel and NGDP targeting

From equation (12) we can note that the wage is a function that depends positively on the future real wage and on the mark-up over wages ($\varepsilon_w/(\varepsilon_w - 1)$), while it depends negatively on the consumer price level (CPI inflation). This relationship is important in order to understand one result of the model: the expectation of an increase in the general price level of the economy causes a fall in the real wage and in the mark-up charged, so that workers will demand a higher optimal wage, generating a higher rate of wage inflation.

On the other hand, equation (18), which describes the dynamics of the evolution of optimal prices, indicates that these depend positively on the output of the economy, on the mark-up over marginal costs ($\varepsilon_p/(\varepsilon_p - 1)$), on the future price level and on the marginal costs of the firm. Were there no wage rigidities, this would imply that domestic inflation and the output of the economy have a direct relationship, so that stabilising domestic inflation would be equivalent to stabilising production (and hence the output gap).

However, since wage rigidities are present, these affect the firm sector through marginal costs. As explained above, a fluctuation in CPI inflation is transmitted directly to the labour market, with a fall in real wages and in the wage mark-up, acting as an endogenous cost shock that will lead to demands for a nominal increase, generating wage inflation, which ends up affecting firms through marginal costs, lowering production, but at the same time raising domestic inflation through the cost channel already mentioned. With lower production, wages cannot be increased while the same labour force is retained, so that employment declines.

This dynamic arising from price and wage rigidities makes IT-PPI an inefficient rule, since it does not respond to the variations present in the labour market; IT-CPI, however, does respond, thereby reducing the endogenous cost shock. This justifies the decision of central banks that in practice consider the CPI when assessing their monetary policy. But, unlike the CPI rule, NGDP also takes into account the real side of the economy, so that it stabilises not only inflation but also output, which in passing allows the labour market to be stabilised, lowering wage inflation. In that sense, the ability of NGDP to attenuate different parts of the economy makes it an efficient policy alternative at the theoretical level. In what follows, we evaluate whether the described properties of NGDP can be replicated in a quantitative assessment.

4 Methodology

4.1 Model calibration

The results of the paper require a calibration of the parameters of the model. The calibrated parameters are presented in Table 2. These are values related to the endogenous dynamics of the model. We exclude only those related to the monetary policy rules, given that these are estimated in order to obtain the optimal rules. Since the main objective of the paper is a general analysis of the welfare properties of NGDP in an open economy, we proceed first with a calibration based on the standard parameters in the literature.

Table 2: Calibrated parameters

Parameter	Value	Name
σ	1	Risk aversion
η	1	Domestic–foreign substitution
γ	1	Substitution among foreign goods
φ	3	Inverse Frisch elasticity
ε_p	6	Elasticity of substitution (goods)
ε_w	6	Elasticity of substitution (labour)
θ_p	[0, 0.75]	Calvo parameter, prices
θ_w	[0, 0.75]	Calvo parameter, wages
β	0.99	Discount factor
α	0.7	Home bias
ρ_A	0.99	TFP autocorrelation
ρ_Z	0.6	Preference autocorrelation
σ_A	0.0071	Standard deviation of the technology process
σ_Z	0.0289	Standard deviation of the preference process

For the subsequent robustness analysis of the results, the model is estimated with values of certain parameters that vary within a specific range. In this way we seek to determine whether the normative properties of a rule are maintained over and above the others, or whether they are parameter specific. To that end, greater importance is given to the parameters governing substitution between domestic and foreign goods and to the degree of openness. In addition, the case of different values of the Calvo parameters (wages and prices) is considered, in a manner similar to that of Garín et al. (2016).

4.2 Welfare computation

For the computation of optimal policy rules and the subsequent welfare analysis, it is necessary to explain the methodology of welfare computation in non-linear models. In the literature there are two types of method for the computation of optimal monetary rules and of welfare. The most common is the analytical approach. Through log-linearisations and specific parameters chosen as assumptions, one arrives analytically at a specific welfare formula, by means of the

LQ method as in Campolmi (2014) for an open economy with price and wage rigidities:

$$\mathbf{V} = -\frac{(1-\alpha)}{2} \left[\frac{\varepsilon_p}{\lambda} \text{var}(\pi_{H,t}) + \frac{\varepsilon_w}{\lambda_w} \text{var}(\pi_{w,t}) + (1+\varphi) \text{var}(x_t) \right]. \quad (26)$$

From this welfare equation the optimal policy rule is derived analytically. This method has two disadvantages. First, the analytical computation of the welfare function only considers specific cases of parameters, and does not allow more general cases to be analysed given the complexity inherent to each of the different cases. Second, in log-linearising the variables, their second moments are lost.

To address these limitations, the literature makes use of second-order approximations through the perturbation method, which allows the first and second moments of the variables to be obtained. This computational method makes it possible to obtain an approximation to the welfare function for different parameters. The variables considered in the computation are the same as in the analytical case. In our case it will depend on domestic inflation, the growth rate of the nominal wage and the output gap, but with a different functional form, which is approximated. The computed welfare is re-expressed in terms of consumption utility, in order to allow comparisons (Adjemian et al. 2022; Born and Pfeifer 2020; Schmitt-Grohé and Uribe 2006):

$$E_0 \left[\sum_{t=0}^{\infty} \beta^t U(C_t, N_t) \right] = E_0 \left[\sum_{t=0}^{\infty} \beta^t U \left((1-\omega_c) C_t^{flex}, N_t^{flex} \right) \right]. \quad (27)$$

The left-hand side of (27) represents utility in the case in which nominal rigidities are present. The right-hand side represents utility without rigidities (the deterministic steady state), but discounted by $1 - \omega_c$, where ω_c is the cost, or the welfare loss, associated with a specific model.

In addition to the method described above, the policy equations are also evaluated against other welfare loss criteria, with the aim of providing greater clarity about the results. One of these loss functions is equation (26), which is the loss function derived from the model itself and depends on domestic inflation, wage inflation and the output gap. Finally, we also consider the following criterion from Woodford (2003), which is standard in the monetary policy literature:

$$L = (\pi_t - \bar{\pi})^2 + \lambda^* x_t^2. \quad (28)$$

4.3 Optimal rules

On the basis of the methodology for welfare computation, we proceed to determine the optimal rules. This consists of obtaining the optimal parameters for each monetary rule, that is, those parameters that reduce the welfare loss. For this purpose we proceed with a recursive algorithm (a simple explanation is provided in Appendix A). With these parameters we run the simulations and compare the welfare losses in the results section.

4.4 Impulse response dynamics

Before turning to the welfare rankings, it is useful to inspect the dynamics that each rule induces. Figure 1 reports the impulse response functions (IRFs) of four models: the Taylor rule, inflation targeting, CPI inflation targeting (IT-CPI) and NGDP targeting. The horizontal axis of the simulations shows periods as quarters, and the vertical axis shows deviations in percentage points with respect to the equilibrium value. The IRFs for the preference and external demand shocks are reported in Appendix B.

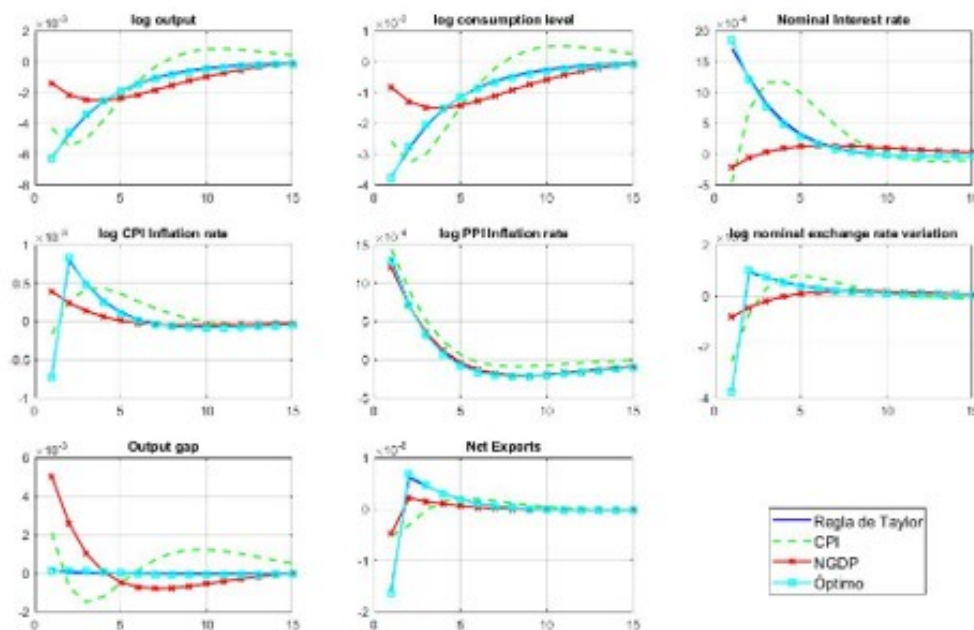


Figure 1: Productivity shock

As shown in Figure 1, in the face of a negative productivity shock the Taylor policy rules display a larger reduction in output, whereas the NGDP rule does not display so large a reduction. Output also falls initially under the remaining policy rules. The behaviour of consumption and of output is very similar. Interest rates are positive or negative at the outset depending on the focus of the rule. As regards the optimal rule and the Taylor rule, both display an increase because the output gap and PPI inflation rise. On the other hand, NGDP and IT-CPI cut rates in order to offset the fall in output and in CPI inflation, respectively. IT-CPI lowers rates aggressively in order to stabilise inflation fully (which it achieves), while NGDP converges more smoothly towards equilibrium because it weighs the fall in real output against the rise in prices. An interesting result is that NGDP, by maintaining stability in production, comes to promote a level of output above the potential level, so that the output gap is larger than under the remaining rules.

The nominal exchange rate declines under the three classical rules. This means that, faced with an appreciation, less domestic currency is required per unit of foreign currency, exports become more expensive and imports cheaper, affecting the level of consumption of households

still further. On the other hand, NGDP brings about a smaller fall in the nominal exchange rate owing to the mild reduction in interest rates.

A positive preference shock is equivalent to an increase in the velocity of money in the economy, boosting production, consumption and domestic inflation. Accordingly, output rises in all scenarios except under the optimal rule. In order to control the effect of the shock, the policymaker withdraws money supply, which is equivalent to raising the interest rate. It should be noted that the scale of the variations in response to the preference shock is relatively smaller than the scale of the variations observed under a productivity shock, because all the rules manage to stabilise this change in the economy adequately. IT-CPI is the rule that, given its concern to avoid a fall in the price level, allows some inflation, which generates a scenario in which output grows more at the outset. The NGDP rule smooths and sustains output growth for a larger number of quarters. The Taylor rule also displays growth, but, like IT-CPI, tends to reduce output rapidly as the interest rate rises.

In the face of an external demand shock, foreign demand for local production increases, so that net exports rise, which implies an inflow of foreign currency into the local economy and appreciates the exchange rate. In the model it can be seen that there is a wealth effect, since the higher foreign demand makes workers and firms wealthier, so that they cease to work or produce as they used to in equilibrium; but this does not affect consumption, which expands as a result of the increase in income. Given that the exchange rate appreciates, and that there is higher demand for domestic goods, local firms can produce and export a smaller quantity and still generate revenue because of the increase in prices (the appreciation of the exchange rate). On the other hand, the appreciation allows households to consume more of what is produced abroad, while at the same time it lowers CPI inflation.

With the exception of IT-CPI, all the rules are qualitatively similar under this shock. They attempt to make the fall in output as small as possible, for which they generate small variations in the interest rate. IT-CPI, by contrast, seeks to maintain stability in the level of consumer prices, so that, faced with the fall arising from the appreciation of the exchange rate, the central bank under this rule will decide to lower the interest rate aggressively, which brings about a larger fall in production than in the remaining cases. In this way CPI inflation returns more rapidly to its equilibrium, and at the same time the exchange rate returns to equilibrium more quickly. These variations cause consumption, which was initially positive, to display a sharp fall. With respect to CPI inflation, all the rules display a nominal deflation of the basket of local and foreign goods, followed by growth and a break towards equilibrium, while the CPI moves in a positive, smoothed and undulating manner towards its equilibrium. The story is different, however, for domestic inflation alone, since it increases in all scenarios. The optimal rule, the Taylor rule and NGDP smooth this inflation through an undulation towards equilibrium around the fifth quarter.

Finally, in a cost shock scenario the intuition is that the marginal costs of the economy rise and the mark-up increases, which is passed through to prices and generates inefficiency in the allocation of resources. As shown in Figure 2, there is an increase in inflation, while the output gap declines; that is, firms produce below the efficient optimal level. There are differences in

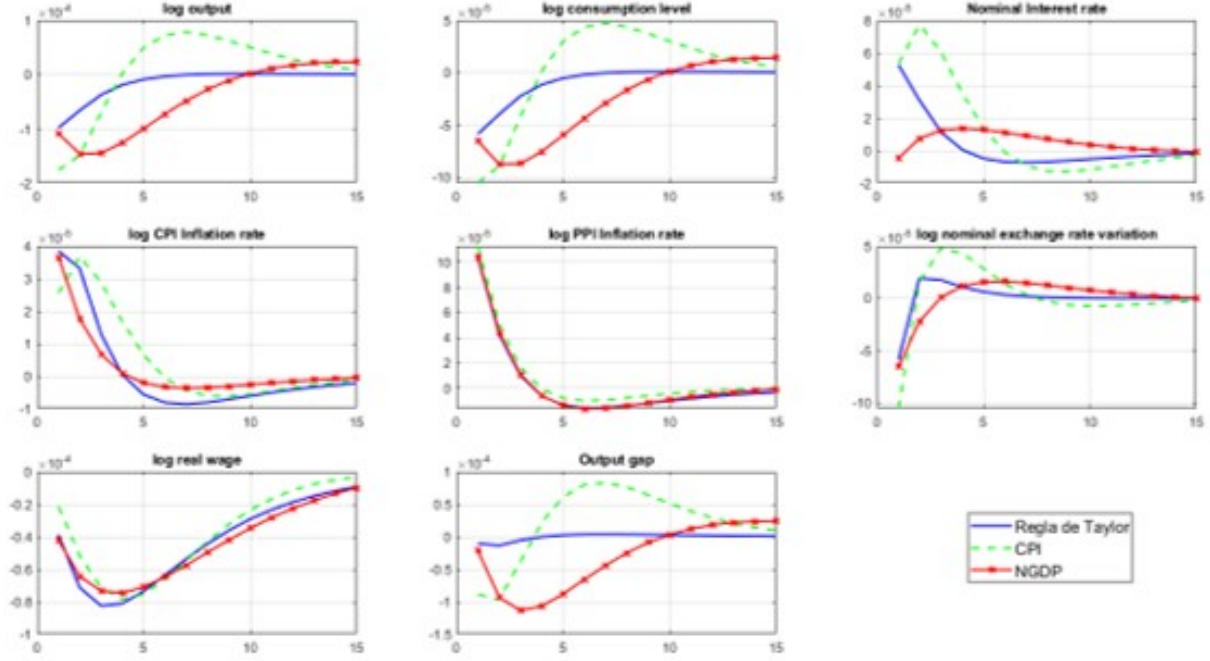


Figure 2: Cost shock

the responses of the monetary policy instrument. The Taylor rule and IT-CPI raise the interest rate, whereas NGDP lowers the policy instrument slightly and then increases it.

5 Analysis of results

On the basis of the monetary policy rules described above, we now present the comparison across models on the basis of the computation of the welfare loss for different specifications of price and wage rigidities.

5.1 Supply shocks

Negative supply shocks play an important role, being the worst scenarios in that they affect the productive capacity of an economy. It is therefore relevant to evaluate the policy rules in the face of this shock. Table 3 presents the results for the optimal rules for domestic inflation (PPI), headline inflation (CPI) and NGDP. In each case the preference for one rule or another varies. Note the case of $\theta_w = 0.75$ and $\theta_p = 0.75$: the results break with the case presented by Galí and Monacelli (2005), since NGDP generates a welfare loss equivalent to 1.02 per cent of consumption, which is the smallest in magnitude. It is followed by the optimal CPI rule at 1.70 per cent, and lastly by the optimal PPI rule (or inflation targeting) at 2.33 per cent of consumption.

It is relevant to note a pattern in the results. Once wage rigidities are included, NGDP turns out to be the optimal policy rule in almost all cases. When $\theta_p = 0.75$ and $\theta_w = 0.25$, its welfare

Table 3: Welfare loss under a productivity shock (per cent of consumption)

θ_p	0	0.75	0.5	0.25	0.75	0.5	0.75
θ_w	0.75	0	0.5	0.75	0.25	0.75	0.75
<i>Panel A: numerical method</i>							
PPI	12.50	0.02	5.34	11.58	1.08	6.51	2.33
Strict CPI	2673.00	0.59	3.83	57.30	0.85	13.31	2.10
Taylor	0.02	0.20	0.95	0.67	0.53	0.89	0.66
CPI	12.55	0.55	2.34	7.91	0.68	4.88	1.70
NGDP	2.41	1.53	0.95	1.69	1.53	1.24	1.02
Wage inflation	0.02	0.58	0.75	0.56	0.58	0.75	0.58
Optimal rule	—	—	0.74	0.56	0.47	0.76	0.58
<i>Panel B: Campolmi (2014)</i>							
PPI	110.47	0.01	9.76	8.65	0.46	6.85	0.57
Strict CPI	51355.31	0.58	5.14	48.86	0.58	13.51	0.87
Taylor	42.52	0.19	4.70	2.53	0.23	4.51	0.11
CPI	110.47	0.54	6.15	5.31	0.35	5.92	0.43
NGDP	70.98	1.51	4.10	3.55	0.91	4.65	0.53
Wage inflation	44.01	0.58	4.37	2.45	0.05	4.35	0.12
Optimal rule	—	—	0.74	0.57	0.44	0.76	0.56
<i>Panel C: Woodford (2003)</i>							
PPI	5.25	0.00	0.42	1.24	0.03	0.33	0.05
Strict CPI	2481.25	0.03	0.19	4.57	0.03	0.67	0.06
Taylor	2.03	0.01	0.22	0.71	0.02	0.21	0.03
CPI	5.25	0.03	0.27	1.01	0.02	0.28	0.04
NGDP	3.39	0.07	0.19	0.85	0.05	0.22	0.05
Wage inflation	2.10	0.03	0.21	0.68	0.03	0.21	0.03
Optimal rule	—	—	0.21	0.71	0.02	0.21	0.03

loss reaches 1.53 per cent of consumption, a figure that is higher (and therefore suboptimal) than the welfare loss under PPI and CPI. In the remaining cases presented, a policy rule based on NGDP is the relevant solution for the model. It is worth mentioning that the second best according to the results turns out to be the CPI inflation target, and that PPI inflation targets turn out to be the most efficient only when wages are flexible ($\theta_w = 0$).

The results regarding the performance of PPI and CPI rules are not particularly new, since they have been theorised and demonstrated by Erceg et al. (2000) and Campolmi (2014). The novel result is that, upon adding an NGDP monetary policy rule to the open economy model with wage rigidities, we find that it generates a smaller welfare loss in the presence of rigidities.

Figure 3 allows us to observe how the performance of the two leading rules varies as the rigidity parameters vary. In general, we observe that NGDP tends to accommodate supply shocks better when wage rigidity is greater than price rigidity. This result arises because inflation directly affects the evolution of wages: if workers expect a rise, they will demand an increase, which affects the firm, mainly when wages are highly rigid and it cannot operate optimally, so that the welfare loss increases. Note that the vertical scales of the two panels differ: at the corner where both rigidities are high, the welfare loss under CPI targeting reaches values

several times above those attained under NGDP targeting.

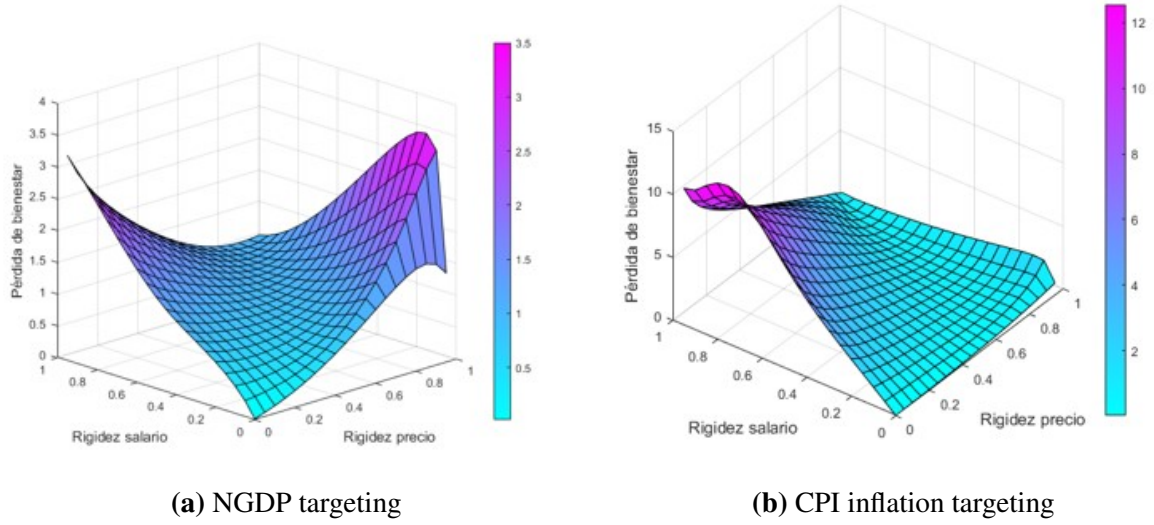


Figure 3: Nominal rigidities and welfare under the two leading rules

Another way of visualising the results shown is through Figure 4, in which we compare the different values for which the welfare generated by NGDP in the face of supply shocks is higher than the welfare generated by the CPI rule.

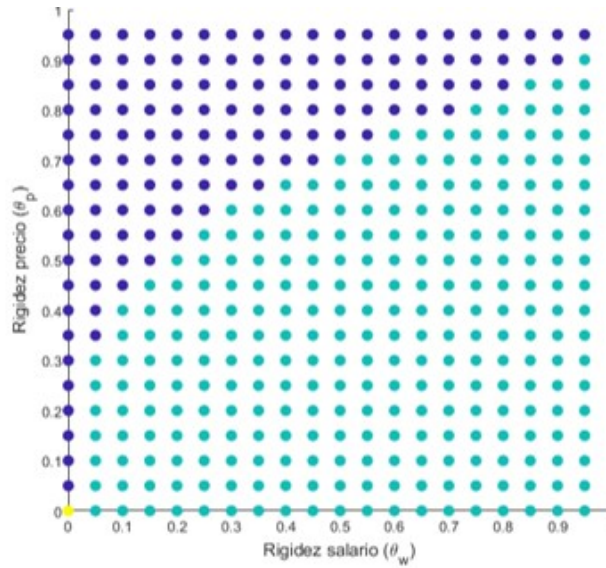


Figure 4: Nominal GDP versus CPI inflation target

Note: the area highlighted in light blue shows the combination of rigidity values (θ_p, θ_w) for which NGDP targeting dominates CPI inflation targeting in terms of generating a smaller welfare loss. The space of dark blue points represents the opposite case. The yellow point reflects the scenario without rigidities, in which the welfare under both rules is the same.

An interesting result is the one shown in Table 4, since central banks can do little or nothing to manage a cost shock, because they must choose lower inflation together with a fall in output. The NGDP rule, under numerical methods, achieves the best performance when the levels of wage rigidity increase, whereas when they decline and there is greater price rigidity there are

scenarios in which PPI ends up first in the ranking of rules that are applicable in practice. When $\theta_p = 0.75$ and $\theta_w = 0.75$, the welfare loss shows the results mentioned in favour of NGDP, while $\theta_p = 0.75$ and $\theta_w = 0.00$ is the scenario in which PPI ends up best.

Table 4: Welfare loss under a cost shock (per cent of consumption)

θ_p	0	0.75	0.5	0.25	0.75	0.5	0.75
θ_w	0.75	0	0.5	0.75	0.25	0.75	0.75
<i>Panel A: numerical method</i>							
PPI	0.0226	0.0006	0.0149	0.0183	0.0073	0.0116	0.0074
Strict CPI	11.5675	0.0006	0.0217	0.1720	0.0061	0.0529	0.0174
Taylor	0.0000	0.0007	0.0027	0.0017	0.0037	0.0028	0.0035
CPI	0.0226	0.0007	0.0083	0.0131	0.0046	0.0099	0.0066
NGDP	0.0130	0.0010	0.0053	0.0093	0.0032	0.0070	0.0051
Wage inflation	0.0002	0.0030	0.0023	0.0015	0.0030	0.0023	0.0030
<i>Panel B: Woodford (2003)</i>							
PPI	0.01527	0.00001	0.00118	0.00299	0.00015	0.00092	0.00022
Strict CPI	3.86291	0.00002	0.00088	0.01427	0.00008	0.00270	0.00045
Taylor	0.00486	0.00003	0.00064	0.00181	0.00013	0.00064	0.00014
CPI	0.01527	0.00002	0.00083	0.00278	0.00011	0.00088	0.00019
NGDP	0.01023	0.00004	0.00062	0.00271	0.00008	0.00086	0.00018
Wage inflation	0.00505	0.00014	0.00061	0.00171	0.00014	0.00061	0.00014

Likewise, under the welfare computation of Panel B, the rules behave in a similar manner, since the ranking indicates that NGDP is the best performing rule when both rigidities are present. On the other hand, the lower the degree of rigidity, the better PPI performs; that is, CPI lies at an intermediate level. Considering that inflation and the output gap are both taken into account, NGDP has better performance under both approaches.

5.2 Demand shocks

We next consider a demand shock, specifically a positive preference shock. This may be understood as a tax cut or a government stimulus. It leads to an increase in consumption; firms anticipate the shock and hire more staff or expand operations. According to the theory set out above, and in line with the impulse response simulations, the results on the welfare loss should be relatively similar. Table 6 in Appendix C reports the results of the model simulations under the shock described.

The results indicate that the three rules are equivalent in the sense that they stabilise the shock optimally and significantly reduce any welfare loss. When both types of rigidity are included, PPI monetary policy becomes less effective as wage rigidity increases. However, the results are effectively maintained between the CPI target rule and NGDP. In both cases the shock is minimised almost entirely and in a similar manner.

Taken together with the supply shock results, this points to two interesting facts. First, the choice of a monetary policy is more relevant in the face of supply shocks than in the face of

demand shocks. Second, NGDP emerges as an efficient alternative when an economy faces only supply shocks and when wage rigidities are not very small.

Considering next a positive external demand shock, the results under numerical methods indicate that PPI monetary policy becomes less effective as wage rigidity increases, but is very effective when prices are made flexible. There is also a similarity with the NGDP rule when both display high price and wage rigidities. NGDP manages to minimise the shock almost entirely and displays a ranking that ends up displacing CPI under scenarios of wage and price rigidities. A similar story arises when the comparison is made with the strict CPI rule. Moreover, under the alternative welfare criteria the same behaviour of the rules is repeated, which confirms the performance of NGDP. When $\theta_p = 0.75$ and $\theta_w = 0.75$, the welfare loss shows the results mentioned in favour of NGDP, with a consumption loss of 0.00003 per cent, whereas when $\theta_p = 0.75$ and $\theta_w = 0.00$ we obtain the scenario in which both PPI and NGDP end up optimal, at 0.00000 per cent. The full set of figures is reported in Table 7 in Appendix C.

5.3 Extension: commodity-exporting emerging economies

Finally, we carry out an additional exercise in order to evaluate the usefulness of the NGDP rule in developing economies that export commodities. To this end, we add an exogenous commodity sector and a government that obtains revenue from the rents of that sector. We evaluate the effect of a rise in the price level of commodities. For this particular exercise, the model is calibrated to the characteristics of a representative commodity-exporting emerging small open economy, drawing on the parameters reported by Castillo et al. (2009); the impulse responses and the additional equations of the model are available upon request.

In general, it can be seen that the shock generates an increase in government spending that promotes higher economic activity: exports increase, the exchange rate appreciates and the current account turns negative after the first quarter. These dynamics are similar to those found for the Chilean case by Medina and Soto (2005).

The simulation results in Table 5 indicate that NGDP continues to be a rule preferable to CPI provided that the economy displays wage rigidities at least as large as its price rigidities. This result is maintained even under the other welfare loss functions. When $\theta_p = 0.75$ and $\theta_w = 0.75$, the welfare loss shows the results mentioned in favour of NGDP, with a consumption loss of 1.0206 per cent, whereas when $\theta_p = 0.75$ and $\theta_w = 0.00$ we obtain the scenario in which PPI is superior to NGDP, ending up optimal with a welfare loss of 0.4299 per cent.

5.4 Robustness tests

In this section we study how the results found previously vary under different calibrations of the model. For this purpose we analyse the persistence of the shocks, the elasticity of substitution between local and foreign goods, the degree of home bias and the Frisch elasticity. The four exercises are collected in Figure 5.

Table 5: Welfare loss under a positive commodity price shock (per cent of consumption)

θ_p	0	0.75	0.5	0.25	0.75	0.5	0.75
θ_w	0.75	0	0.5	0.75	0.25	0.75	0.75
<i>Panel A: numerical method</i>							
PPI	0.4298	0.4299	1.4683	2.5324	0.7346	2.0668	1.6702
Strict CPI	0.4297	2.2002	1.8963	1.3613	4.7881	2.3212	4.6386
Taylor	0.7615	1.1149	0.8141	0.8035	1.1734	0.8751	0.9975
CPI	0.4298	1.2876	1.1486	1.1795	1.7310	2.2499	3.4419
NGDP	1.5952	0.4679	0.8681	1.8036	0.5379	1.4734	1.0206
Wage inflation	0.4298	0.6370	0.5616	0.4953	0.6370	0.5599	0.6372
Optimal rule	—	—	0.5650	0.4962	0.5669	0.5635	0.6457
<i>Panel B: Campolmi (2014)</i>							
PPI	4.39	0.11	1.57	2.28	0.45	2.46	0.94
Strict CPI	4.39	1.75	1.95	0.89	3.83	1.78	2.41
Taylor	4.17	0.72	1.10	0.48	0.49	1.01	0.26
CPI	4.39	0.89	1.10	0.86	1.03	1.71	1.76
NGDP	8.77	0.13	0.81	1.29	0.15	1.36	0.33
Wage inflation	4.39	0.28	0.91	0.41	0.07	0.89	0.10
Optimal rule	—	—	0.23	0.13	0.21	0.20	0.27
<i>Panel C: Woodford (2003)</i>							
PPI	0.21	0.01	0.07	0.29	0.01	0.12	0.06
Strict CPI	0.21	0.08	0.08	0.08	0.08	0.09	0.13
Taylor	0.20	0.03	0.05	0.08	0.02	0.05	0.02
CPI	0.21	0.04	0.04	0.12	0.04	0.08	0.09
NGDP	0.42	0.01	0.03	0.16	0.01	0.07	0.02
Wage inflation	0.21	0.01	0.04	0.09	0.01	0.04	0.01
Optimal rule	—	—	0.04	0.09	0.01	0.04	0.01

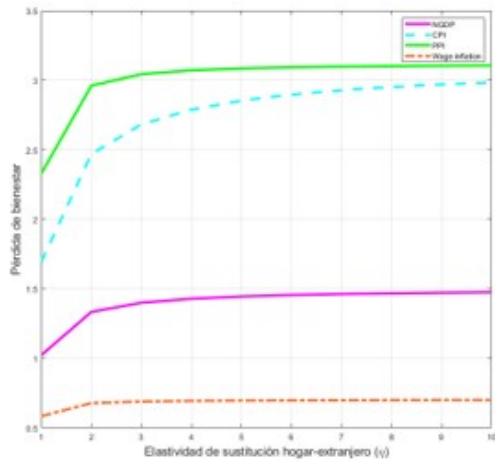
Panel a presents values for the computed welfare loss for different values of the elasticity between local and foreign goods. It can be noted that, among the alternatives modelled, a rule targeting wage inflation is the optimal one. It is followed by NGDP, IT-CPI and IT-PPI, respectively. As the elasticity increases the ranking does not change, but the magnitude of the difference between NGDP and CPI does increase.

With regard to the persistence of the shocks, in panel b it is striking that NGDP is not very effective when these last a single period, or are not very persistent. By contrast, when the persistence coefficient is 0.5, it begins to outperform IT-CPI and IT-PPI, which become progressively less effective. For its part, a wage inflation rule is always the most convenient.

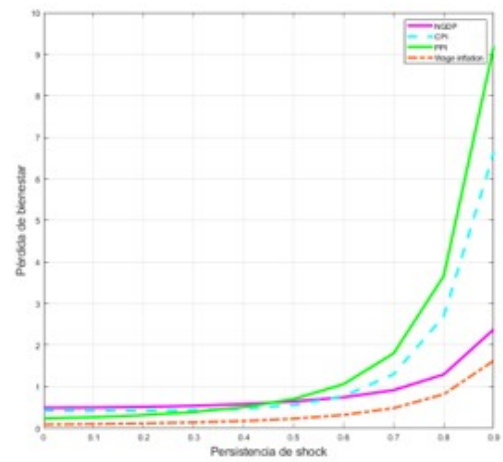
With regard to the degree of home bias, panel c shows interesting results when comparing IT-CPI and NGDP. When there is a high degree of openness to foreign trade, both rules deliver almost similar results. However, to the extent that the economy is more closed, this gap in efficiency between the two rules widens.

Finally, panel d evaluates how the elasticity of labour supply matters. Reducing the Frisch elasticity (equivalent to increasing φ) amplifies the distance between the rules. This may occur because, as the elasticity declines, the effect transmitted in the labour market is amplified in the

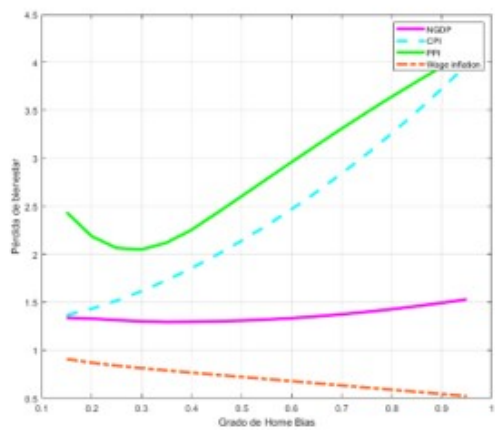
firm sector as a result of the existence of rigidities in the economy.



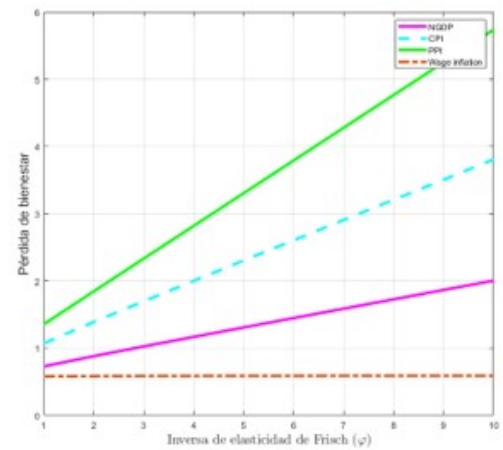
(a) Elasticity of substitution in foreign trade



(b) Persistence of the productivity shock



(c) Degree of home bias



(d) Inverse of the Frisch elasticity

Figure 5: Robustness of the welfare ranking to alternative calibrations

5.5 Limitations of the research

The previous results provide a valid justification in favour of our hypothesis regarding the positive properties of an NGDP rule, in comparison with other simple and feasible rules. There are, however, certain limitations of the work that must be mentioned with respect to the evaluation.

As is common to every DSGE model, a threat and therefore a direct limitation is that the results analysed depend on the description and construction of the model. In that sense, we consider that, for the purposes of the paper, the specification presented is more than sufficient to test the validity of our hypothesis, since what we had in mind was above all a quantitative study. However, if there is greater interest in evaluating the applicability of this rule to a particular economy, it will be necessary to use a more complex model tailored to the scenario one wishes

to represent.

An additional limitation is that this model did not consider a rising trend in either inflation or the growth of the economy. This does not allow us to evaluate one of the most common criticisms made of NGDP, namely the existence of variations or breaks in the long-run growth trend of economies, a phenomenon characteristic of emerging countries.

Finally, the model used does not consider the factor of noise in information; that is, we do not analyse the magnitude of the welfare loss generated by following policy rules that make use of variables with measurement problems, as is mainly the case of the output gap, but also of nominal GDP, as the literature suggests.

6 Conclusions and recommendations

This paper developed and simulated a DSGE model of a small open economy for different monetary policy rules derived from the Taylor rule and for an alternative NGDP rule. Through simulations using numerical methods that allowed the welfare loss to be computed, it was shown that the NGDP targeting rule is the one that reacts best to negative productivity shocks in the majority of the scenarios analysed when wage and price rigidities are present. In addition, the results of the paper found that, for the scenarios studied, the performance of NGDP targeting is similar to that of the other Taylor rules in the face of demand shocks; that is, there are no significant differences.

The results build on the literature on optimal monetary policy, and are similar to previous work. Of the three classical monetary policy rules (Taylor, CPI, PPI), we obtain a result similar to that of Campolmi (2014), in which the optimal CPI rule performs better; and, in line with the work of Garín et al. (2016), in a closed economy NGDP targeting delivers a lower welfare loss relative to the other monetary policy alternatives. Our extension shows that in a small open economy the NGDP targeting rule can be used to obtain the smallest welfare loss.

The fact that NGDP targeting displays a degree of effectiveness in confronting supply shocks relative to the classical alternatives is a relevant result for the discussion of economic policy. This is because of the difficulty a central bank faces in reacting to supply shocks. This fact becomes even more relevant when one considers that, going forward, as a result of recurrent technological and climate-related changes, central banks will face supply shocks more frequently.

The results imply that NGDP should be an alternative to be taken into account; there are, however, costs associated with switching from one policy rule to another, such as the loss of confidence and the costs that households must bear in the face of new rules of the game. In that sense, an interesting extension to the work presented is to run simulations that allow the losses associated with switching from one monetary policy to another to be computed within the model, so that further evidence is gathered on the viability of NGDP as an alternative for the future. On the other hand, the model employed shows the versatility of NGDP targeting, which makes it an interesting alternative to evaluate in scenarios in which traditional monetary policies are not very effective, as is the case of the zero lower bound.

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A Algorithm to obtain the optimal rule

- (a) Determine a vector with initial values and a range of possible values for the parameters of the monetary rule, which are used for the simulation:

$$[\phi_i] \in [1.1; x]; \quad \phi_i = 1.1.$$

- (b) By means of the perturbation method, the level of the welfare loss under the rule is obtained for the first value set, and stored in the vector $\mathbf{W}_c$:

$$E_0 \left[\sum_{t=0}^{\infty} \beta^t U(C_t(\phi_i), N_t(\phi_i)) \right] = E_0 \left[\sum_{t=0}^{\infty} \beta^t U((1 - \omega_c)C_t^{flex}, N_t^{flex}) \right] \implies \mathbf{W}_c = \omega_c(i).$$

- (c) We then proceed to test the next value within the range of values established in step (a).
- (d) The model is solved again as in step (b). The process is repeated while approaching the maximum of the function.
- (e) The value of the parameter that delivers the smallest welfare loss is chosen: a global maximum.

$$\phi_{i, \text{optimal}} = \phi_i(\min\{W_c\}).$$

B Additional impulse response functions

This appendix collects the impulse response functions for the two shocks whose dynamics are discussed in Section 4 but which are not central to the welfare ranking. Figure 6 reports the responses to a positive preference shock and Figure 7 those to a positive external demand shock. In both cases the horizontal axis measures quarters and the vertical axis deviations in percentage points from the equilibrium value.

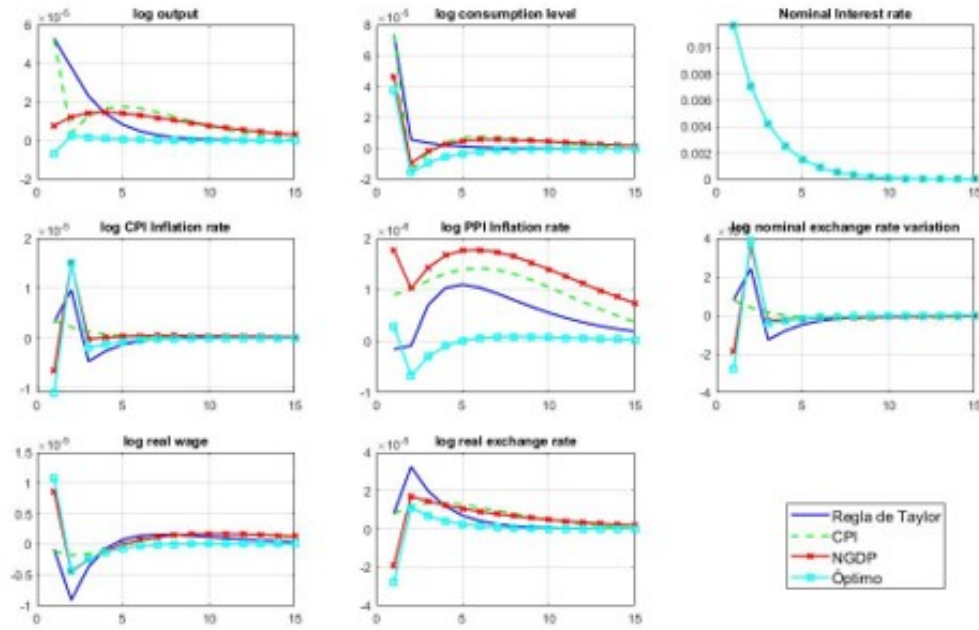


Figure 6: Positive preference shock

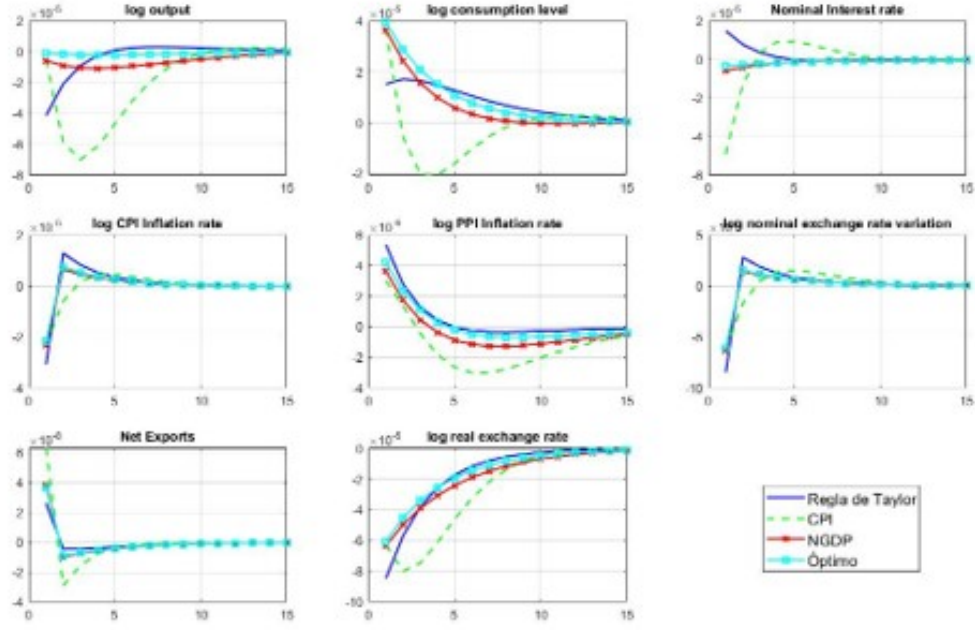


Figure 7: Positive external demand shock

C Additional welfare results

This appendix reports the welfare losses under the two demand-side shocks. Because the policy rules are close to equivalent in these cases, and the losses involved are several orders of magnitude smaller than under supply shocks, the tables are collected here rather than in the main text.

Table 6: Welfare loss under a demand shock (per cent of consumption)

θ_p	0	0.75	0.5	0.25	0.75	0.5	0.75
θ_w	0.75	0	0.5	0.75	0.25	0.75	0.75
<i>Panel A: numerical method</i>							
PPI	-0.00553	-0.00554	3.22797	15.28394	0.20527	10.86399	9.34009
Strict CPI	-0.00567	-0.00566	-0.00566	-0.00566	-0.00564	-0.00566	-0.00566
Taylor	-0.00546	-0.00562	-0.00547	-0.00546	-0.00549	-0.00546	-0.00546
CPI	-0.00553	-0.00555	-0.00552	-0.00558	-0.00554	-0.00556	-0.00558
NGDP	-0.00559	-0.00558	-0.00558	-0.00563	-0.00557	-0.00563	-0.00562
Wage inflation	-0.00553	-0.00556	-0.00559	-0.00562	-0.00558	-0.00562	-0.00562
Optimal rule	—	—	-0.00568	-0.00567	-0.00568	-0.00568	-0.00568
<i>Panel B: Campolmi (2014)</i>							
PPI	0.00018	0.00014	3.02300	15.27800	0.22251	10.99400	8.93600
Strict CPI	0.00014	0.00002	0.00002	0.00002	0.00004	0.00002	0.00002
Taylor	0.00008	0.00003	0.00004	0.00000	0.00006	0.00000	0.00000
CPI	0.00018	0.00014	0.00014	0.00006	0.00012	0.00007	0.00006
NGDP	0.00027	0.00011	0.00010	0.00003	0.00006	0.00003	0.00003
Wage inflation	0.00018	0.00014	0.00007	0.00005	0.00005	0.00005	0.00004
Optimal rule	—	—	0.00000	0.00000	0.00000	0.00000	0.00000
<i>Panel C: Woodford (2003)</i>							
PPI	0.000009	0.000007	0.136233	1.181927	0.005812	0.568510	0.462866
Strict CPI	0.000007	0.000001	0.000001	0.000001	0.000001	0.000001	0.000001
Taylor	0.000004	0.000002	0.000001	0.000001	0.000002	0.000000	0.000000
CPI	0.000009	0.000007	0.000004	0.000004	0.000003	0.000003	0.000003
NGDP	0.000012	0.000005	0.000003	0.000002	0.000003	0.000001	0.000001
Wage inflation	0.000009	0.000007	0.000002	0.000003	0.000003	0.000002	0.000002
Optimal rule	—	—	0.000000	0.000001	0.000000	0.000000	0.000000

Table 7: Welfare loss under an external demand shock (per cent of consumption)

θ_p	0	0.75	0.5	0.25	0.75	0.5	0.75
θ_w	0.75	0	0.5	0.75	0.25	0.75	0.75
<i>Panel A: numerical method</i>							
PPI	0.00004	0.00000	0.00008	0.00012	0.00003	0.00007	0.00003
Strict CPI	0.00000	0.00014	0.00015	0.00010	0.00041	0.00017	0.00027
Taylor	0.00000	0.00002	0.00002	0.00001	0.00002	0.00001	0.00001
CPI	0.00004	0.00008	0.00010	0.00016	0.00017	0.00025	0.00042
NGDP	0.00010	0.00000	0.00003	0.00007	0.00001	0.00005	0.00003
Wage inflation	0.00004	0.00001	0.00002	0.00004	0.00001	0.00003	0.00002
Optimal rule	—	—	0.00001	0.00001	0.00001	0.00001	0.00001
<i>Panel B: Campolmi (2014)</i>							
PPI	0.000680	0.000004	0.000116	0.000124	0.000031	0.000102	0.000021
Strict CPI	0.000479	0.000142	0.000159	0.000091	0.000406	0.000148	0.000198
Taylor	0.000423	0.000018	0.000056	0.000025	0.000009	0.000046	0.000004
CPI	0.000680	0.000083	0.000105	0.000135	0.000134	0.000214	0.000220
NGDP	0.001037	0.000004	0.000054	0.000083	0.000006	0.000080	0.000012
Wage inflation	0.000680	0.000007	0.000052	0.000048	0.000002	0.000060	0.000008
Optimal rule	0.000000	0.000000	0.000009	0.000006	0.000006	0.000008	0.000006
<i>Panel C: Woodford (2003)</i>							
PPI	0.0000325	0.0000002	0.0000052	0.0000177	0.0000006	0.0000050	0.0000014
Strict CPI	0.0000228	0.0000069	0.0000064	0.0000065	0.0000064	0.0000076	0.0000100
Taylor	0.0000202	0.0000009	0.0000026	0.0000068	0.0000005	0.0000022	0.0000004
CPI	0.0000325	0.0000040	0.0000040	0.0000168	0.0000038	0.0000107	0.0000106
NGDP	0.0000497	0.0000002	0.0000023	0.0000131	0.0000003	0.0000039	0.0000007
Wage inflation	0.0000325	0.0000003	0.0000023	0.0000097	0.0000003	0.0000029	0.0000006
Optimal rule	0.0000000	0.0000000	0.0000023	0.0000069	0.0000003	0.0000021	0.0000003